OpenDRIVE® 1.8.0, Static Road Network Description, 2023-11-22

Introduction

In ASAM OpenDRIVE, signals have the @country attribute to differentiate between the signals and signs from different countries. As a default set of signals the country `OpenDRIVE` is available. How to use and define signals is defined in Section 14.1, "Introduction to signals".

A signal with the @type and @subtype attributes is only unique in combination with the @country and @countryRevision attributes. Since some elements that are considered signals in ASAM OpenDRIVE, for example traffic lights, do not have any official @type and @subtype representation, these are specified in this document. They can be used with the appropriate @type, @subtype and the @country="OpenDRIVE".

XML example

```
<signals>
   <signal        s="3981.4158159146"
                  t="-14.0503"
                  id="5000162"
                  name="Vorschriftzeichen"
                  dynamic="no"
                  orientation="+"
                  zOffset="3.8835"
                  country="DE"
                  countryRevision="2017"
                  type="274"
                  subtype="100"
                  value="100"
                  unit="km/h"
                  height="0.77"
                  width="0.77"
                  hOffset="5.7595865"/>
</signals>
```

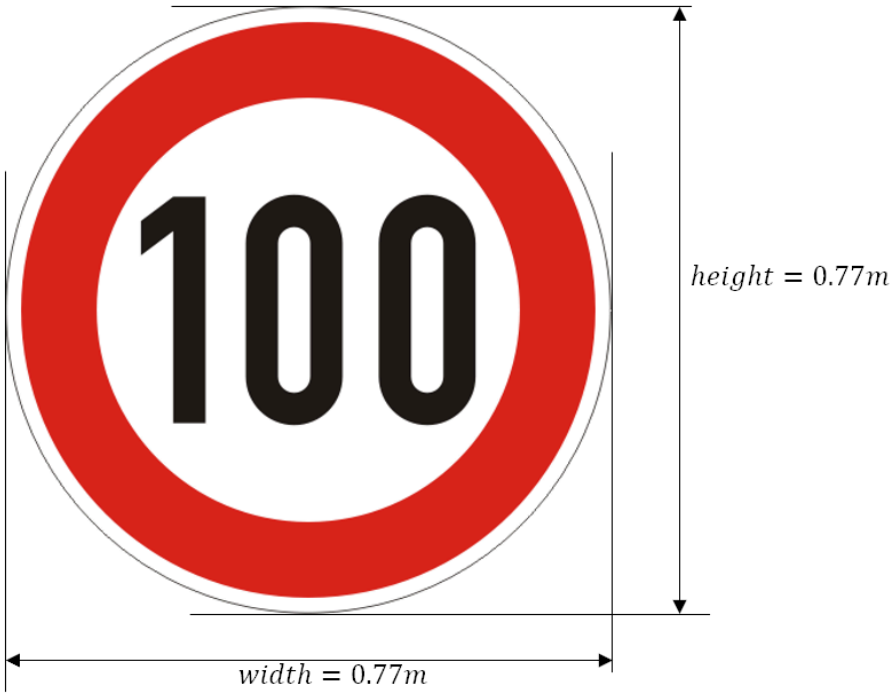


Figure 1. Example for a signal definition

In ASAM OpenDRIVE 1.5 the @countryRevision attribute was introduced. This is due to the fact that, with a new revision of the signal definition, done by the legal authorities, some types and subtypes of signals have changed. For example, in the country revision from 2017 the speed limit sign has the following values for @type and @subtype:

```
type = 274
subtype = 50
```

Table 1. Example for a signal with country revision 2017

Country revision	Sign	Country	Type	Subtype	Unit	Value

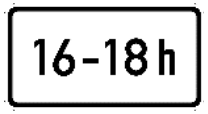
Country revision	Sign	Country	Type	Subtype	Unit	Value
2017		DE	274	50	km/h	50

In the former revision from 2013 the following values for @type and @subtype were set:

```
type = 274
subtype = 55
```

Table 2. Examples for signals with country revision 2013

Country revision	Sign	Country	Type	Subtype	Unit	Value
2013		DE	274	55	km/h	50
2013		DE	265	-1	m	3,5
2013		DE	110	-1	%	10

Country revision	Sign	Country	Type	Subtype	Unit	Value
2013		DE	1040	-1	m	16001800

The most recent German sign definition can be found at *VzKat.de* [1].

Additional information regarding the German traffic signs can be found in the book *HAV - Hinweise für das Anbringen von Verkehrszeichen und Verkehrseinrichtungen* [2].

The United States traffic signs can be found here:

- *Standard Highway Signs - 2012 Supplement to the 2004 Edition* [3] (PDF, main source for use)
- *Standard Highway Signs, 2004 Edition* [4] (PDF)
- *Manual on Uniform Traffic Control Devices for Streets and Highways* [5] (HTML)
- *Standard Highway Signs—PDF and EPS files for New and Revised Signs* [6] (HTML)
- *United States Road Symbol Signs* [7] (HTML)
- *United States Road Symbol Signs* [8] (PDF)

Speed signals in the United States do not have unique @type and @subtypes attributes, the speed value is given in the @value attribute and in the @unit attribute.

Table 3. United States speed limit signal example with country revision 2014

Country revision	Sign	Country	Type	Subtype	Unit	Value
2014		US	R2	1	mph	50

Speed limits drawn on the road as road marks are not necessarily signals. In many countries these road marks are not binding and must be accompanied by additional signs.

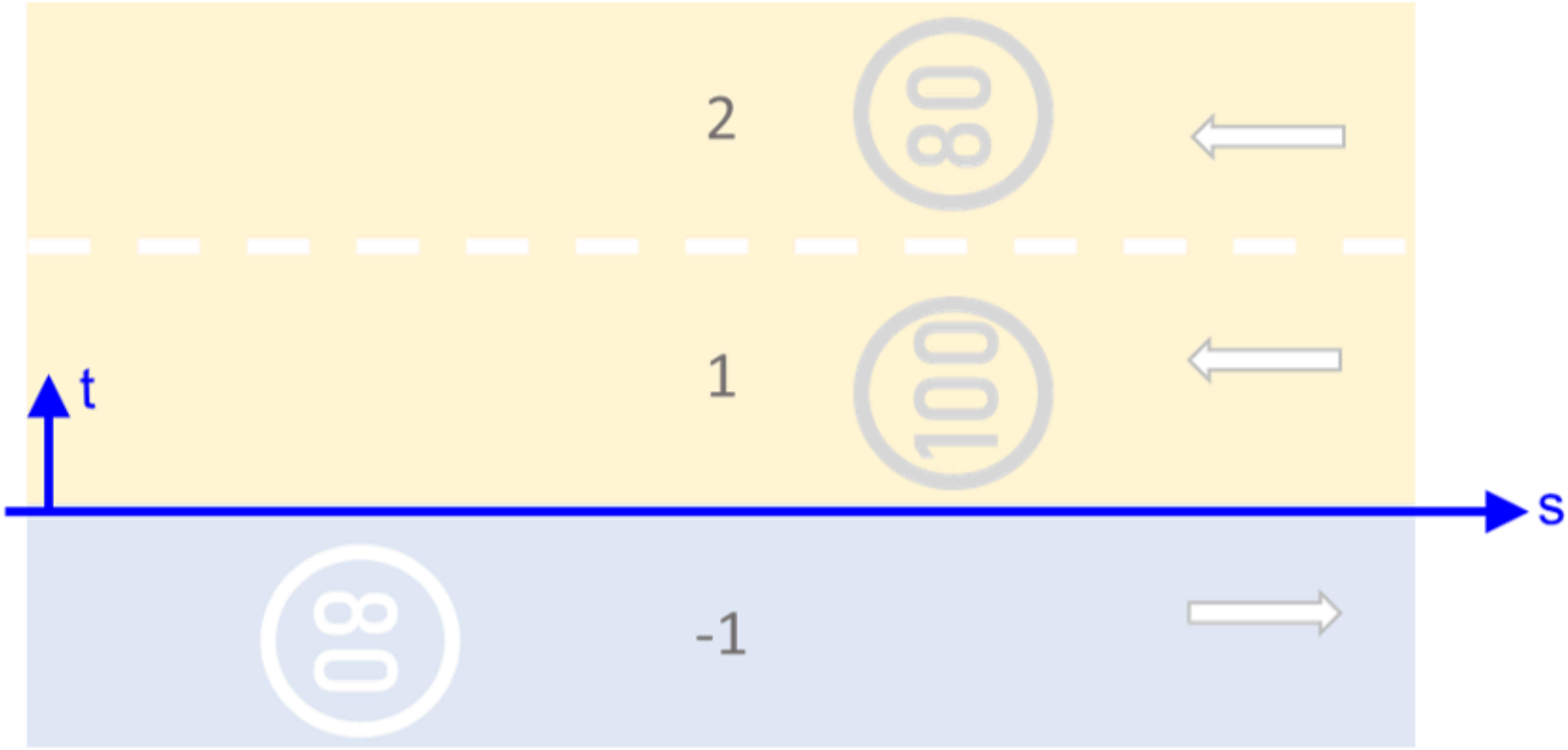


Figure 2. Road marks defining a speed limit

When using the official signal catalogs, provided by the authorities, use the following type and subtype encoding:

```
[type]-[subtype]
If there is no subtype available, set @subtype = `-1`
```

Table 4. Example for subtype=-1

Country revision	Sign	Country	Type	Subtype	Unit	Value	Text

Country revision	Sign	Country	Type	Subtype	Unit	Value	Text
		DE	310	-1			Wilster\n Kreis Steinburg